

# IRexp experimental infrared band lists from the PMC Open Access Subset and Chemotion

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## Abstract

IRexp is a redistributable collection of **experimental infrared band lists** ( $\text{cm}^{-1}$  peak positions) mined from open chemistry literature, optionally with  $^1\text{H}/^{13}\text{C}$  shifts and resolved structures. The release holds **121,233** records (119,345 PMC OA; 1,888 Chemotion/RADAR4Chem), with **43,060** structure-linked and **33,201** full IR +  $^1\text{H}$  +  $^{13}\text{C}$  + structure quadruples. IRexp stores **numeric band lists**, not absorbance traces. Records carry `source_doi` and stamped `license / license_pool`. Licensing is **mixed**: Europe PMC join yields **87,617** commercial (CC-BY/CC0), **20,938** non-commercial (NC\*), **10,781** empty/unknown (excluded from commercial Zenodo), and **1,897** ShareAlike (Chemotion + rare PMC SA). Reuse: multimodal training and complementary elucidation benchmarks citing this Descriptor. Data: Hugging Face ilkhamfy/IRexp; Zenodo DOI pending archival mint. Code is MIT-licensed.

## 1 Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet **open, redistributable** collections of *experimental* IR data remain sparse relative to modern machine-learning needs. Digitised absorbance libraries such as the NIST Chemistry WebBook [National Institute of Standards and Technology, 2026] and AIST SDBS [National Institute of Advanced Industrial Science and Technology, 2026] are valuable but either modest in size or **view-only** (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra [Zipoli et al., 2025], while large literature mines for **NMR** peak lists (notably NMRexp [Wang et al., 2025]) demonstrate that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans [Yang et al., 2026]) further motivate an **IR-focused, redistributable** band-list resource rather than another closed spectrum archive.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound **IR band lists** (wavenumbers in  $\text{cm}^{-1}$ ) together with  $^1\text{H}/^{13}\text{C}$  NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a **different object** from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the **PMC Open Access Subset** bulk S3 mirror and peak lists from the **Chemotion** FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).

3. Resolves compound names to canonical structures with OPSIN [Lowe et al., 2011], RD-Kit [Landrum, 2026], and SELFIES [Krenn et al., 2020] where possible.
4. Releases **extracted numbers only** — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IReXP is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IReXP holds structure-linked band lists. The intended reuse is multimodal pretraining, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — **not** reproduced here) [Yabbarov and Vargas-Hernández, 2026].

**What this Data Descriptor does not contain.** No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

## 2 Methods

### 2.1 Source corpora

**PMC Open Access Subset.** Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`) [National Library of Medicine, 2026]. **Harvest window (recoverable from git snapshots):** the bulk S3 IR crawl and `seen_papers / ir_harvest_snapshot` artefacts were produced on **2026-06-07 (UTC)** (`scripts/s3_ir_harvest.py`; incremental auto-snapshots that day through ~134,893 raw IR rows). Chemotion was ingested the same day. A harvest snapshot records **188,016** distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from **15,416** unique PMC:\* accessions. Extraction operates on open-access **plain text** objects at flat S3 keys `PMC{id}.*v*/PMC{id}.*v*.txt`. Europe PMC full-text XML is used for some validation re-fetches. The **released** IReXP DOIs are exclusively PMC:\* accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

**Discovery vs `oa_comm` / `oa_noncomm` packages.** Identifier discovery used NCBI E-utilities `esearch` over PMC with an IR-characterisation query `and open access[filter]`, then fetched matching IDs from the flat S3 text layout. The harvest therefore did **not** pre-filter by walking the PMC OA Subset’s `oa_comm` vs `oa_noncomm` vs other package directories. PMC OA is still **not** a single licence [National Library of Medicine, 2026]. Licence truth for redistribution is applied **post-hoc**: every unique PMCID in IReXP (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license` / `license_pool`. Commercial-use (CC-BY/CC0) rows form the Zenodo / Sci Data primary pool; NC\* are held aside; empty/unknown are **excluded** from the commercial deposit — aligning redistributable intent with the OA commercial/non-commercial distinction via article-level licences rather than S3 package paths (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

**Chemotion / RADAR4Chem.** **1,888** records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI 10.22000/0GoEQGlsZGEIrgst) [Jung et al., 2024], licensed **CC-BY-SA-4.0**. Ingest (`scripts/chemotion_to_irexp.py`, 2026-06-07): download the MD5-verified deposit; for each of **2,116** ATR-IR analyses, flatten the Quill-delta `content` field to plain text; parse an **author-curated** IR band list with the **same** regex extractor and quality gates as the PMC path; resolve the deposit’s canonical SMILES with RDKit →

InChIKey + SELFIES; keep the richest band list per InChIKey; drop the **8** molecules already present in the PMC pool  $\rightarrow$  **+1,888** new structure-resolved rows. These are **not** algorithmic peak-picks from absorbance curves; they are author-entered experimental band lists from the ELN deposit, denser than typical paper prose (median **39** bands vs **9** for PMC). Rows carry `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI.

**Excluded.** AIST SDBS (view-only; no bulk export) [National Institute of Advanced Industrial Science and Technology, 2026]; NIST WebBook join code exists in the repository but contributes **0** records to the released IReXP (`ir_source` is always `experimental` for released rows).

## 2.2 Discovery and fetch (released corpus)

1. Discover IR-reporting OA PMCIDs via NCBI `esearch` (`open access[filter]` + characterisation-format IR query; year/month sliced).
2. Fetch OA full text from PMC-OA S3 plain-text objects (`PMC{id}.{v}.txt`).
3. Ingest Chemotion deposit author-curated band lists + structures (`scripts/chemotion_to_irexp.py`).
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl.gz` (134,893 rows including intermediate prose fields; the curated release is `irexp.jsonl.gz`).

**Non-OA / Scraping fence.** Development adapters for ChemRxiv, Beilstein, and generic publisher pages exist for exploration. They are **outside the construction path of the released dataset**. **No released `source_doi` is a paywalled publisher DOI**. Methods for IReXP as published here cite only **PMC-OA S3 + Chemotion**.

## 2.3 Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers  $\rightarrow$  `ir_bands_cm-1` (list of floats,  $\text{cm}^{-1}$ );
- $^1\text{H}$  and  $^{13}\text{C}$  payloads  $\rightarrow$  `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives. **No curve digitisation** is performed: figures are not traced; JCAMP-DX is not stored.

## 2.4 Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN [Lowe et al., 2011], canonicalised with RDKit [Landrum, 2026], and encoded as InChIKey and SELFIES [Krenn et al., 2020]. An optional PubChem [Kim et al., 2023] name fallback (`USE_PUBCHEM`) handles trivial names. Structure coverage of the full release is **35.5%** (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

## 2.5 Licence handling

| Pool                                 | Records       | Stamp                                                           |
|--------------------------------------|---------------|-----------------------------------------------------------------|
| commercial (CC-BY + CC0)             | <b>87,617</b> | <code>license_pool=commercial;</code> Zenodo / Sci Data primary |
| non_commercial (CC-BY-NC*)           | 20,938        | held aside                                                      |
| sharealike (Chemotion + rare PMC SA) | 1,897         | CC-BY-SA / CC-BY-SA-4.0                                         |
| empty_unknown                        | 10,781        | excluded from commercial deposit                                |

`scripts/join_pmc_licences.py` stamps every row; `scripts/split_license_pools.py` reports provenance and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

## 2.6 Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` ( $^{13}\text{C}$  peak count  $\leq$  carbon count;  $^1\text{H}$  integration vs formula; IR wavenumber windows). They were **not** applied as a hard filter at harvest; Technical Validation reports a full-corpus post-hoc quarantine on `irexp_resolved` (`scripts/quarantine_structure_nmr.py`  $\rightarrow$  `data/audit/structure_nmr_quarantine.jsonl.gz`). Transcription and recall-proxy scripts: `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`.

## 3 Data Records

### 3.1 Object definition

Each IReXP record is a JSON object. Required chemistry fields for an IR entry:

| Field                                                              | Type           | Description                                              |
|--------------------------------------------------------------------|----------------|----------------------------------------------------------|
| <code>id</code>                                                    | string         | Stable record id (InChIKey when resolved, else internal) |
| <code>ir_bands_cm-1</code>                                         | float list     | Experimental IR peak positions ( $\text{cm}^{-1}$ )      |
| <code>ir_source</code>                                             | string         | <b>experimental</b> for all released rows                |
| <code>source_doi</code>                                            | string         | PMC:<pmcid> or Chemotion deposit DOI                     |
| <code>h_nmr</code> / <code>c_nmr</code>                            | string or null | Author-reported shift lists                              |
| <code>smiles</code> / <code>selfies</code> / <code>inchikey</code> | string or null | Resolved structure encodings                             |
| <code>has_structure</code>                                         | bool           | Convenience flag                                         |
| <code>license</code> / <code>license_pool</code>                   | string         | Per-article stamp (Europe PMC join or Chemotion)         |

**Not included:** absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

### 3.2 Files and counts

Paths relative to the project repository / Hugging Face mirror.

| File                                               | Records       | Description                                       |
|----------------------------------------------------|---------------|---------------------------------------------------|
| data/irexp/irexp.jsonl.gz                          | 121,233       | Full curated release                              |
| data/irexp_resolved/<br>irexp_resolved.jsonl.gz    | 43,060        | 100% structure-linked                             |
| data/irexp_release/<br>train_no_bench.jsonl.gz     | 42,808        | Resolved minus IRSpectra-Bench<br>InChIKeys       |
| data/irexp_release/<br>train_no_bench_nmr.jsonl.gz | 32,949        | Same with both <sup>1</sup> H and <sup>13</sup> C |
| data/irexp_release/<br>pretrain_ir.jsonl.gz        | 119,345       | PMC-only IR pretrain pool                         |
| data/irexp/seen_papers.txt.gz                      | 188,016 lines | PMC IDs scanned at harvest                        |
| data/irexp/<br>ir_harvest_snapshot.jsonl.gz        | 134,893       | Intermediate harvest snapshot                     |
| licence_pools/<br>irexp_commercial.jsonl.gz        | 87,617        | CC-BY + CC0 (Zenodo/Sci Data<br>primary)          |
| licence_pools/<br>irexp_non_commercial.jsonl.gz    | 20,938        | NC* held aside                                    |
| licence_pools/<br>irexp_sharealike.jsonl.gz        | 1,897         | Chemotion + rare PMC SA                           |
| licence_pools/<br>irexp_empty_unknown.jsonl.gz     | 10,781        | Excluded from commercial                          |

### Composition of irexp.jsonl.gz.

| Slice                                                  | Count          |
|--------------------------------------------------------|----------------|
| All IR band-list records                               | 121,233        |
| With <sup>1</sup> H and/or <sup>13</sup> C NMR         | 87,075 (72%)   |
| With resolved structure                                | 43,060 (35.5%) |
| Structure + any NMR                                    | 40,702         |
| Full IR + <sup>1</sup> H + <sup>13</sup> C + structure | 33,201         |
| PMC OA provenance                                      | 119,345        |
| Chemotion provenance                                   | 1,888          |
| Unique PMC accessions                                  | 15,416         |

**Licence-pool counts (confirmed Track 1 join, 2026-08-26).** Lookup over **15,416** unique PMCID; stamped pool files under data/irexp/licence\_pools/. Sum of pools = **121,233** (unchanged).

| Pool file                     | Count         | Notes                                                           |
|-------------------------------|---------------|-----------------------------------------------------------------|
| irexp_commercial.jsonl.gz     | <b>87,617</b> | CC-BY / CC0 — <b>Zenodo primary</b>                             |
| irexp_non_commercial.jsonl.gz | <b>20,938</b> | CC-BY-NC* held aside                                            |
| irexp_empty_unknown.jsonl.gz  | <b>10,781</b> | empty / unresolved — <b>excluded</b> from<br>commercial deposit |
| irexp_other.jsonl.gz          | <b>0</b>      | reserved (e.g. CC-BY-ND)                                        |
| irexp_sharealike.jsonl.gz     | <b>1,897</b>  | Chemotion CC-BY-SA-4.0 (1,888) +<br>rare PMC SA                 |

The full irexp.jsonl.gz remains multi-licence on disk; commercial redistribution must use the commercial pool (or license\_pool == "commercial"). Hugging Face was re-issued with stamped pools and honest card text (scripts/publish\_hf.py, 2026-08-26).

Median bands: **9** (PMC), **39** (Chemotion). All **1,360,866** released IR band values fall inside 350–4000 cm<sup>-1</sup> (full-corpus range check; Technical Validation).

### 3.3 Access

- **Hugging Face:** <https://huggingface.co/datasets/ilkhamfy/IRexp>
- **GitHub development mirror:** <https://github.com/IlkhamFY/spectro-agent>
- **Zenodo archival snapshot:** DOI pending mint after honest multi-licence metadata.

## 4 Technical Validation

Machine-readable package: `docs/scientific_data/qc_structure_nmr.json` (and artefacts under `data/audit/`).

### 4.1 Transcription fidelity (PMC)

On a seed-fixed random sample of **60** PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ( $\pm 1\text{ cm}^{-1}$  integer match). Result: **560/560 bands** and **60/60 records** confirmed. Enlarged sample **n=200** (same script/seed family, `data/audit/extraction_audit_n200.json`): **2,250/2,261 bands (99.51%)** and **196/200 records (98.0%)** fully confirmed. This bounds **transcription** error (hallucinated, mis-parsed, or unit-mangled values).

### 4.2 Extraction-recall proxy (PMC, harvest path)

A human mark-up of every IR string in every paper remains the gold standard. As an automatic **proxy** on the actual harvest path (`scripts/audit_extraction_recall.py --n 40 --seed 0`): re-fetch PMC-OA S3 plain text for **40** distinct source PMCID; re-run `extract_records`; compare band-sets to released rows ( $\pm 1\text{ cm}^{-1}$ ). Result: **2,640/2,640** released bands confirmed in re-extract; **248/248** released IR lists recovered (list-level recall proxy **1.0**; 40/40 papers); **4/40** papers yielded *extra* re-extracted lists relative to the curated release (parser over-fire vs post-harvest dedup/curation, not missed released compounds). Summary: `data/audit/extraction_recall_proxy.json`. This is **not** a substitute for expert human recall.

### 4.3 Structure-NMR consistency and quarantine (resolved)

**Sample (prior).** On **500** `irexp_resolved` records with  $^{13}\text{C}$  text (seed 0): **17/500 (3.4%)** listed more peaks than carbons. On **500** with  $^1\text{H}$  text: integrals  $>$  formula  $\text{H}+2$  in **17/497 (3.4%)**.

**Full resolved corpus.** `scripts/quarantine_structure_nmr.py` applied the same physics gates to all **43,060** structure-linked rows. **1,882 (4.37%)** fail  $\geq 1$  hard check and are listed in `data/audit/structure_nmr_quarantine.jsonl.gz` (diagnostic only — release files unchanged). Among rows with the relevant modality:  $^{13}\text{C}$  peaks  $>$  carbons **1,194/34,231 (3.49%)**;  $^1\text{H}$  integral  $>$  formula+2 **1,141/39,672 (2.88%)**; IR out-of-range **0**; unparseable SMILES **0**. Re-users should filter quarantined IDs before supervised training. These rates are integrity diagnostics, **not** an expert skeleton audit.

### 4.4 IR physical window

Every band in the full 121,233-record release lies in **[350, 4000]  $\text{cm}^{-1}$**  (**0** / 1,360,866 out of range).

## 4.5 QC status

| Check                                                    | Status                                    |
|----------------------------------------------------------|-------------------------------------------|
| Per-PMCID licence join + pool counts                     | <b>Done</b> — commercial 87,617           |
| Transcription fidelity n=60 and n=200                    | <b>Done</b>                               |
| Extraction-recall automatic proxy (n=40 papers, S3 path) | <b>Done</b> (human recall still optional) |
| Full-corpus structure-NMR quarantine                     | <b>Done</b> — 1,882 / 43,060 flagged      |
| Expert structure spot-check ( $n \geq 100$ )             | Optional / deferred                       |

## 5 Usage Notes

- **Band lists  $\neq$  spectra.** Do not evaluate models trained on IRExp as if they had seen full absorbance curves.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, author-curated ELN band lists). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi / pmcid`.
- **Structure-NMR quarantine.** Before supervised training on `irexp_resolved`, optionally drop IDs in `data/audit/structure_nmr_quarantine.jsonl.gz` ( $\sim 4.4\%$  of resolved rows).
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems, fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record’s `source_doi`.

## 6 Data Availability

IRExp numeric extracts are available at:

- Hugging Face Datasets: <https://huggingface.co/datasets/ilkhamfy/IRExp>
- GitHub: <https://github.com/IlkhamFY/spectro-agent> (data/irexp/, data/irexp\_resolved/, data/irexp\_release/)
- Zenodo: archival DOI at proof; primary artifact = commercial pool, cc-by-4.0 metadata + SA companion  
Licensing summary (honest):
- **Chemotion (1,888):** CC-BY-SA-4.0 [Jung et al., 2024].
- **PMC (119,345):** mixed Creative Commons — stamped per article; commercial redistributable **87,617** (CC-BY/CC0); NC\* **20,938** held aside; empty/unknown **10,781** excluded from commercial Zenodo (LICENCE\_REMEDIATION.md).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

## 7 Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in <https://github.com/IlkhamFY/spectro-agent> under the **MIT License** (LICENSE). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); release harvest `scripts/s3_ir_harvest.py`, `scripts/chemotion_to_irexp.py`;

validation     `scripts/audit_extraction.py`,     `scripts/audit_extraction_recall.py`,  
`scripts/quarantine_structure_nmr.py`; pool split `scripts/split_license_pools.py`.  
The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

## 8 Author contributions

**I.Y.:** conceptualization, methodology, software, data curation, validation, writing — original draft.

**R.A.V.-H.:** conceptualization, supervision, writing — review and editing.

## 9 Competing interests

The authors declare no competing interests.

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